

ESF: Harmonizing Heterogeneous EEG Acquisition for Pretrained Classifiers

Hitesh Dammu

Independent Researcher, Hyderabad, India

hdammu@asu.edu

Abstract—Pretrained EEG classifiers are trained on one acquisition setup and deployed on many. We ask a narrow question: given a fixed, deterministic preprocessing contract, which acquisition shifts does it close, for which backbone, and at what cost? The contract is the **Encephlian Standard Format (ESF)**: 19-channel 10–20 at 250 Hz, notch and high-pass filtered, average-referenced, robust z-scored; every stage is standard. We apply five operator-modelled shifts (native rate, gain, mains, montage, broadband noise) to raw microvolt TUAB evaluation recordings ($n=276$, patient-disjoint from probe training) and compare the full contract against the contract with the corrective stage disabled, for three frozen public backbones (EEGPT, LaBraM, BIOT) and four probes each. Gain and mains are recovered exactly by every backbone (naive losses up to 0.30 AUROC). Native rate is recovered at every rate for EEGPT but not at 128 Hz for BIOT ($0.90 \rightarrow 0.67$) or LaBraM ($0.89 \rightarrow 0.82$); a low-pass sweep shows why: EEGPT’s ranking is unchanged down to 15 Hz, while BIOT’s 64 Hz cell reproduces its 128 Hz cell without resampling. Recoverability is a property of the (shift, backbone) pair, and so is the direction of failure: uncorrected shift turns normal recordings into confident abnormal reports for EEGPT and BIOT, and abnormal ones into confident normal reports for LaBraM under mains interference. The contract removes these failures where it is measured to, and nowhere else. Permuted-label controls, a failed label-free monitor and six recordings from a 125 Hz Natus recorder complete the protocol. v2 corrects v1 (Version note). Code, operators, probes, caches and per-recording predictions are released.

Index Terms—EEG, canonical representation, acquisition shift, pretrained models, recoverability, selective prediction.

I. INTRODUCTION

A pretrained EEG classifier is trained once, on recordings from one instrumentation pipeline, and then asked to read whatever a clinic owns: different sampling rates, gain conventions, electrode subsets, regional mains, often legacy hardware. Self-supervised backbones [1–3] report strong held-out accuracy on the Temple University corpora [4, 5], on data from the pipeline that produced their training corpus. Existing robustness benchmarks corrupt the model input directly; none simulates the recorder and asks what a preprocessing contract can undo.

This paper is a measurement protocol for such a contract, applied to three backbones the author did not train (Fig. 1). The contract is the **Encephlian Standard Format (ESF)**, whose stages are all standard preprocessing; its content is that they are fixed, ordered and applied identically at training and deployment. Five acquisition axes are simulated as physical operators on the raw microvolt recording at native rate and, for

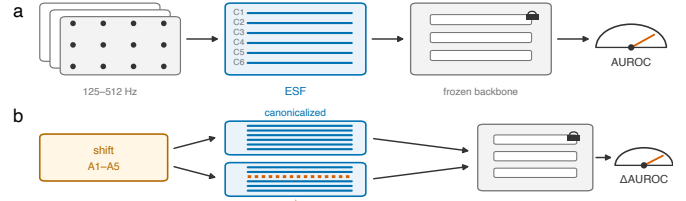


Fig. 1: **Deployment and protocol.** (a) Recorders that differ in rate, channel set, gain, mains and export feed one deterministic six-stage contract (ESF) and one frozen backbone and probe; nothing downstream is retrained. (b) An acquisition operator acts on the raw microvolt recording; the canonicalized arm applies the full contract, the naive arm the same contract with the one stage that addresses the axis switched off; both feed the same frozen backbone, so their difference is the recovery attributable to that stage.

each axis, the one contract stage that addresses it is switched off, so the recovery attributable to that stage is measured under matched pairing. Contributions: (i) the protocol and its operators, released; (ii) a recoverability framework whose left-invertibility half predicts which axes close before any data is seen, whose Shannon half is a conditional the data can test, and whose AUROC–total-variation bound is stated class-conditionally; (iii) EEGPT, LaBraM and BIOT measured with three MLP seeds and a linear probe each; (iv) a per-backbone deployment envelope (Table 1), a bandwidth measurement that explains the axis on which backbones disagree, and a per-recording failure taxonomy; (v) negative controls, a label-free monitor that fails, and a real legacy recorder.

II. BACKGROUND

BIOT tokenizes heterogeneous biosignals [3], LaBraM scales masked-token pretraining to 2,500 h [2], EEGPT adds representation alignment [1]; linear probing of a frozen backbone is the standard evaluation [6]. The in-corpus to deployment gap defines clinical machine learning [7–9]; ImageNet-C [10] set the parameterized perturbation methodology adapted here. BIDS-EEG [11] and NWB [12] standardize datasets; ESF standardizes the signal a model sees. Three contemporaneous lines work from the model side: BRIDGE-EEG [13] maps recordings to a device-agnostic 62-channel time–frequency representation before pretraining its own encoders; PRISM [14] pretrains across acquisition systems; graph montage harmonization [15] learns a mapping between electrode layouts. None simulates the recorder as operators,

holds a fixed contract on the other side, ablates it stage by stage, or measures third-party backbones (Table S7).

III. METHODS

A. Data

Probes are trained on the TUAB v3.0.1 *train* split [5, 16] (2,717 recordings, 2,076 patients; patient-stratified 90/10 train/validation, seed 7) and evaluated once on the official *eval* split ($n=276$, 253 patients, 150 normal, 126 abnormal; patient-ID intersection with train 0). No evaluation recording touches training, validation, calibration or the fixed-scale constants of the naive gain arm.

B. Backbones, adapters and probes

Three frozen public checkpoints via brain-decode 1.5.2: EEGPT (eegpt-pretrained), LaBraM (labram-pretrained) and BIOT (biot-pretrained-prest-16chs). Each consumes ESF through a fixed model-side adapter applied identically to every arm and cell: EEGPT, 19 channels at 250 Hz in 4 s windows, encoder tokens averaged to 512-d; LaBraM, 19 named channels at 200 Hz in its pretrained 15 s windows, patch tokens averaged to 200-d; BIOT, the 16 longitudinal bipolar derivations formed from the 19 referential channels at 200 Hz in 5 s windows, 256-d encoder embedding. EEGPT’s public checkpoint lacks its 19→19 channel projection; it is initialised from seed 7, frozen and released. Each backbone carries four probes on the pooled per-recording representation (mean, max, 95th percentile of ℓ_2 -normalised window vectors): three MLP probes ($3d \rightarrow 256 \rightarrow 1$, dropout 0.2, seeds 7/11/13, AdamW 10^{-3} , class-weighted BCE, 25 epochs, best validation AUROC) and one logistic probe; train-statistics standardisation, Platt scaling on validation. Windows are pooled in up to eight groups; the recording score is the mean group logit and the across-group variance is kept as a label-free monitoring statistic.

C. Pretraining provenance

Evaluation recordings must be unseen at every stage, including pretraining. EEGPT was pretrained on non-TUH data (PhysioNet motor imagery, the Tsinghua SSVEP benchmark, M3CV and SEED, per the authors’ repository). LaBraM’s pre-training includes TUAR, TUEP, TUSZ and TUSL but excludes TUAB and TUEV, recording-disjointly; patient-level overlap with TUAB is not reported and is a residual, unquantified risk for LaBraM only. For BIOT we use the checkpoint pretrained on Massachusetts General Hospital resting EEG alone; the six-dataset checkpoint, which includes TUAB train, is not used.

D. The contract

ESF v1.0: **(C1)** the 19 international 10–20 sites [17] in fixed order; **(C2)** 250 Hz by anti-aliased polyphase resampling; **(C3)** physical microvolts from the EDF header, then channel-wise robust z-score (median, IQR, persisted); **(C4)** common average reference over good channels; **(C5)** IIR notch at the regional mains and harmonics, and a 0.5 Hz high-pass; **(C6)** missing

channels are NaN with a quality flag, never imputed. Reference implementation: MIT.

E. Acquisition-shift axes

Every operator A_θ acts on the *raw* microvolt recording at native rate, before any contract stage (Fig. 1b; Table S6). The *canonicalized* arm then runs C1–C6; the *naive* arm runs the same pipeline with the one corrective stage disabled, so the arms differ only in that stage and share every downstream weight. (A1) native rate $f_n \in \{128, 200, 250, 256, 512\}$ Hz by anti-aliased resampling; naive arm presents native samples as if 250 Hz. (A2) scalar gain $g \in \{0.5, 0.75, 1, 1.5, 2\}$; naive arm normalises with fixed train-set scale constants. (A3) 60 Hz mains and harmonics below native Nyquist, mean peak $a \in \{0, 5, 15, 30, 60\}$ μ V, per-electrode log-normal amplitude ($\sigma=0.3$) and random phase; naive arm has the notch off. (A4) montage *full19*, *legacy16* (T3, T4, T5, T6, Fz, Cz, Pz absent) or *bipolar18* re-projected to referential slots; naive arm averages over all 19 slots with zeros. (A5) white noise per channel at $\text{SNR} \in \{-5, 0, 5, 10, 20\}$ dB; no corrective stage exists, so the arms coincide. A sixth, measurement-only axis low-passes the raw recording at 15–100 Hz (§IV-B).

F. Recoverability framework

Let A_θ map a canonical signal to its native-device representation and K_θ be a canonicalization operator targeting $K_\theta \circ A_\theta = \text{id}$; f is the frozen probe and $\mathcal{A}(f; \mathcal{D})$ its AUROC. An axis is (f, ε) -recoverable iff some K_θ keeps $|\mathcal{A}(f \circ K_\theta \circ A_\theta) - \mathcal{A}(f)| \leq \varepsilon$ for every θ in the support.

Proposition 1 (Left-invertibility $\Rightarrow$ recoverability). *If A_θ admits a left-inverse within the contract, the axis is $(f, 0)$ -recoverable for every f .*

Proposition 2 (Sub-Nyquist irrecoverability). *If A_θ is an ideal anti-alias low-pass at $f_n/2$ followed by decimation with $f_n < 2f_{\max}$, and f depends on content in $(f_n/2, f_{\max}]$, no K_θ recovers f ’s clean AUROC.*

Theorem 1 (AUROC–TV bound, class-conditional). *For any bounded-score f , with $P|_\pm$ the class-conditional score distributions, $|\mathcal{A}(f; \mathcal{D}_0) - \mathcal{A}(f; \mathcal{D}_\theta)| \leq \text{TV}(P_0|_+, P_\theta|_+) + \text{TV}(P_0|_-, P_\theta|_-) \leq \sum_\pm \sqrt{\frac{1}{2} D_{\text{KL}}(P_\theta|_\pm || P_0|_\pm)}$.*

Proposition 1 applies to gain and mains. Proposition 2 is a statement about the classifier, not the axis. The marginal score distribution, which a deployed system can observe without labels, is *not* bounded this way (v1 stated a marginal form with $C(f) \leq 4$; that step was invalid). Proofs and a composability lemma: Appendix A.

G. Statistics

Canonicalized-vs-naive contrasts use DeLong’s test on matched recordings [18]; intervals are paired class-stratified bootstraps (1,000 resamples); Benjamini–Hochberg at $q=0.05$ over the non-identity cells of each backbone [19]. Tables report the mean over the three MLP seeds; tests use the seed-7 probe.

TABLE 1: **Envelope.** Change in AUROC from each backbone’s clean baseline on TUAB eval ($n=276$), per cell and arm, mean over three MLP seeds; shading is proportional to the loss, saturating at -0.5 . Bold: canonicalized-arm cells more than 0.01 below baseline, i.e. outside the envelope. A5 has no corrective stage, so its naive row is empty. Paired tests per contrast in Table S4.

		A1 rate (Hz)				A2 gain ($\times$)				A3 mains (μ V)				A4 montage		A5 SNR (dB)				
		128	200	256	512	0.5	0.75	1.5	2	5	15	30	60	leg16	bip18	-5	0	5	10	20
EEGPT	naive	-0.17	-0.02	-0.01	-0.17	-0.08	-0.04	-0.02	-0.02	-0.02	-0.10	-0.22	-0.30	-0.01	-0.02					
	canonicalized	0	0	0	0	0	0	0	0	0	0	0	0	-0.01	-0.02	-0.16	-0.11	-0.06	-0.04	-0.01
LaBraM	naive	-0.08	-0.05	-0.01	-0.29	-0.02	-0.02	-0.01	-0.01	-0.03	-0.09	-0.11	-0.21	-0.04	-0.02					
	canonicalized	-0.08	0	0	0	0	0	0	0	0	0	0	0	-0.04	-0.02	-0.20	-0.15	-0.12	-0.09	-0.05
BIOT	naive	-0.32	-0.05	0	-0.49	-0.06	-0.05	-0.02	-0.04	-0.24	-0.27	-0.25	-0.27	-0.03	-0.02					
	canonicalized	-0.23	0	0	0	0	0	0	0	0	0	0	0	-0.03	-0.02	-0.22	-0.11	-0.09	-0.06	-0.05

IV. RESULTS

A. Baselines and per-axis results

Clean canonicalized AUROC, mean over seeds: EEGPT 0.900 (0.895–0.904; balanced accuracy 0.817, ECE 0.082), BIOT 0.898 (0.892–0.901; 0.827, 0.062), LaBraM 0.894 (0.890–0.903; 0.796, 0.111); linear probes 0.88–0.89. Window count carries no label information (AUROC 0.49). Table 1 gives every cell; Fig. 2a its seed spread; Tables S1–S4 balanced accuracy, ECE and tests. (A1) *Rate*. EEGPT recovers every rate: canonicalized 0.900 at each of 128/200/256/512 Hz against naive 0.729/0.879/0.892/0.727 (128 and 512 Hz FDR $p < 10^{-6}$); the only residual is calibration (ECE 0.10 vs. 0.08 at 128 Hz). BIOT recovers 200, 256 and 512 Hz (canonicalized 0.893/0.898/0.898; the 512 Hz naive arm is below chance, 0.403, $p = 4 \times 10^{-36}$) but not 128 Hz: canonicalized 0.668 (0.645–0.692 over seeds), balanced accuracy 0.50, normal recordings pushed to a median logit of +6. Proposition 2’s premise fails for EEGPT and holds for BIOT. LaBraM is a third case: 200 and 512

Hz are recovered (naive 0.845 and 0.608 to canonicalized 0.898 and 0.895, $p = 3$

$\times 10^{-4}$ and 2

$\times 10^{-10}$), but 128

Hz is not, and resampling does nothing for it (naive 0.813, canonicalized 0.815, balanced accuracy 0.50). (A2) *Gain*. Canonicalized AUROC equals baseline at every g for every backbone. The naive arm loses 0.080 (EEGPT) and 0.062 (BIOT) at $g=0.5$ and is below baseline even at $g=1$ (0.878, 0.862); fixed constants are themselves a mis-normalization. LaBraM’s naive arm is already tolerant of gain (losses

≤ 0.02 , none significant). (A3) *Mains*. Canonicalized AUROC equals baseline at every amplitude for every backbone. EEGPT’s naive arm falls to 0.878/0.801/0.683/0.598 at 5/15/30/60 μ V and from 15 μ V its balanced accuracy is 0.50 with ECE above 0.5: it fails open into one class with confidence. BIOT’s collapses at 5 μ V (0.661) and is below baseline with no injected interference (0.862), since notch-off also leaves the corpus’s own residual mains. LaBraM’s naive arm falls to 0.863/0.801/0.786/0.681 and, unlike the other two, pushes

emphabnormal recordings toward normal (S

refsec:failure). (A4) *Montage*. Recovered by no backbone and not worsened by the naive average: EEGPT loses 0.012 (legacy16) and 0.020 (bipolar18) on both arms; BIOT 0.031 and 0.019. LaBraM loses most, 0.044 on texttlegacy16 with balanced accuracy 0.53. The loss is in the acquisition, not the reference. (A5) *Broadband*. Arms identical; AUROC falls monotonically, EEGPT 0.737 $\rightarrow$ 0.890 and BIOT 0.678 $\rightarrow$ 0.847 from -5 to 20 dB. LaBraM 0.691 to 0.847, with balanced accuracy at 0.50 up to 5 dB.

B. Bandwidth is a property of the backbone

Low-passing the raw recording at 15–100 Hz (zero-phase 8th-order Butterworth, before the contract; arms identical) explains the one axis on which backbones disagree (Fig. 2b,c). EEGPT’s ranking is flat, AUROC 0.899–0.903 at every cutoff from 15 to 100 Hz (mean over seeds); its calibration is not, with balanced accuracy 0.81 $\rightarrow$ 0.65 and ECE 0.07 $\rightarrow$ 0.31 from 30 to 15 Hz. BIOT’s curve is neither flat nor monotone: 0.68 at 64 Hz, 0.85 at 30 Hz, 0.59–0.70 at 20 Hz across seeds, with both classes pushed above logit +9 from 20 to 30 Hz. Its 64 Hz cell reproduces its 128 Hz cell (0.68 vs. 0.67) with no resampling; BIOT’s sub-Nyquist failure is the loss of 64–100 Hz content, which its 1 Hz spectrogram tokenizer was pretrained on. The envelope has two edges, one for ranking and a tighter one for calibrated decisions.

C. Failure taxonomy

Across the 12 non-identity cells of A1–A3, and per arm, each recording is assigned one of seven categories from its trajectory (Fig. 3c; Table S5); *catastrophic shift* is baseline-correct, then $p_{\text{wrong}} \geq 0.80$ on at least one cell. On the naive arm, for EEGPT every one of 110 baseline-correct normals flips to a confident abnormal report on some cell while 91 of 114 baseline-correct abnormalities stay correct; for BIOT 124/124 and 96/126. For LaBraM the load falls on both classes: 104 of 119 normals flip at 512 Hz, and 90 of 103 abnormalities flip to a confident *normal* report at 60 μ V mains (67 at 30, 37 at 15 μ V). The direction of failure is therefore a property of the (axis, backbone) pair; for a screening tool the normal-to-abnormal direction is the costly one [20], the other is the dangerous one. Canonicalization removes every confident flip for EEGPT (the 29 “calibration

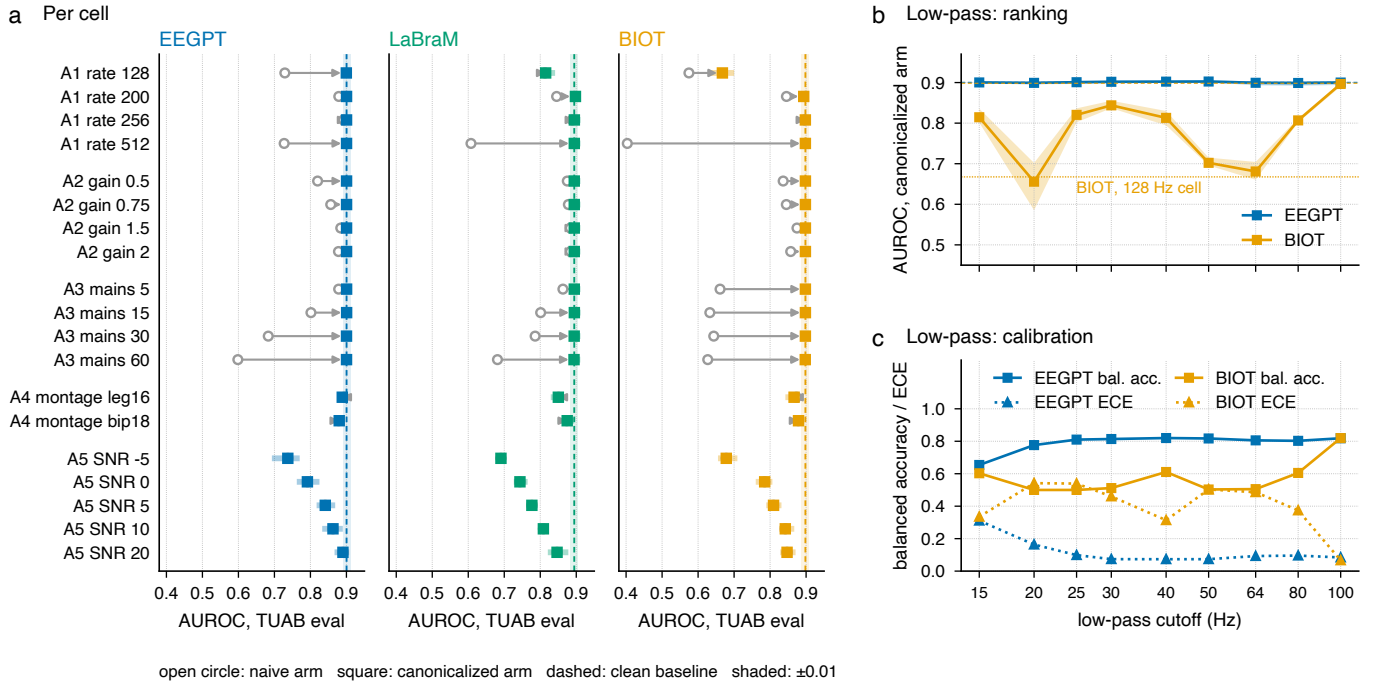


Fig. 2: **Every cell, and why the backbones disagree.** (a) AUROC of the naive arm (open circle) and the canonicalized arm (filled square) per backbone; the arrow is the recovery the stage delivers, the band the min–max over three probe seeds, the shaded strip the ± 0.01 envelope around the clean baseline. (b) AUROC of the canonicalized arm after a low-pass applied to the raw recording: EEGPT’s ranking is unchanged down to 15 Hz; BIOT’s 64 Hz cell reproduces its 128 Hz cell with no resampling. (c) Balanced accuracy and ECE over the same cutoffs.

collapse” recordings have baseline probabilities 0.40–0.49 and are borderline, not shifted), and for BIOT and LaBraM every one except at 128 Hz, where 100 of 124 and 79 of 119 baseline-correct normals flip. The “flips on some cell” statistic is a maximum over 12 cells and is not compared against a matched in-distribution null; it describes this sweep, not a population rate.

D. Controls, monitor, bound

Permuted labels. Ten MLP probes and one linear probe per backbone trained with training and validation labels permuted reach eval AUROC 0.46 ± 0.08 (EEGPT; linear 0.55), 0.48 ± 0.10 (LaBraM; linear 0.51) and 0.58 ± 0.04 (BIOT; linear 0.44). The BIOT residual is unsupervised outlyingness, not leakage: with no labels, mean distance to the ten nearest training recordings separates the classes at 0.74 for BIOT, 0.70 for LaBraM and 0.51 for EEGPT, and untrained random heads score 0.47 ± 0.09 , 0.49 ± 0.13 and 0.53 ± 0.14 . **Label-free monitor.** Neither the across-group logit variance nor $|p - 0.5|$ ranks catastrophic recordings above the rest (median AUROC 0.54 over 11 naive cells; abstaining on the highest-variance 20% removes 24% of them). **Bound.** Over the 40 measured non-identity arms the class-conditional bound holds on 40/40 for EEGPT and 40/40 for BIOT, 40/40 for LaBraM, and is vacuous (sum of class-conditional TVs above 1) on the strongly shifted naive arms; the marginal ratio has median 0.15 (EEGPT), 0.20 (LaBraM) and 0.24 (BIOT) and maximum 1.41, 2.25 and 1.35, so no marginal constant below 1 exists for any backbone in this sweep.

E. A real legacy recorder

Six de-identified routine recordings from a Natus recorder in clinical use (125 Hz, 24–27 channels, 50 Hz mains, proprietary .e export; no reads) were parsed by the released reader, canonicalized and scored locally; all six resolved to the contract. All three probes call all six abnormal (EEGPT at percentiles 0.67–0.97 of its clean eval logits, LaBraM 0.90–1.00, BIOT 0.63–0.94; KS against clean eval $p \leq 0.008$). The recorder lies inside EEGPT’s envelope and outside LaBraM’s and BIOT’s (Fig. 3b), so only EEGPT’s verdict is testable; whether the six are abnormal or residually shifted needs reads, and we do not guess.

V. DISCUSSION

Recoverability belongs to the pair. Gain and mains close for every backbone, as Proposition 1 predicts before any data; native rate closes for EEGPT and not, at 128 Hz, for BIOT or LaBraM, and the low-pass sweep locates BIOT’s dependence in its 64–100 Hz bins. LaBraM fails at 128 Hz too, and there resampling changes nothing, so the loss is in the band, not the grid. A contract measured on one backbone says nothing about another. Montage and broadband noise close for none; the deployable response there is intake-side abstention keyed on logit magnitude, since every probe fails open with confidence and probability-ambiguity rules miss exactly the documented failures.

Contract design. Because EEGPT’s ranking survives a 15 Hz low-pass and its calibration a 30 Hz one, ESF v1.1 will

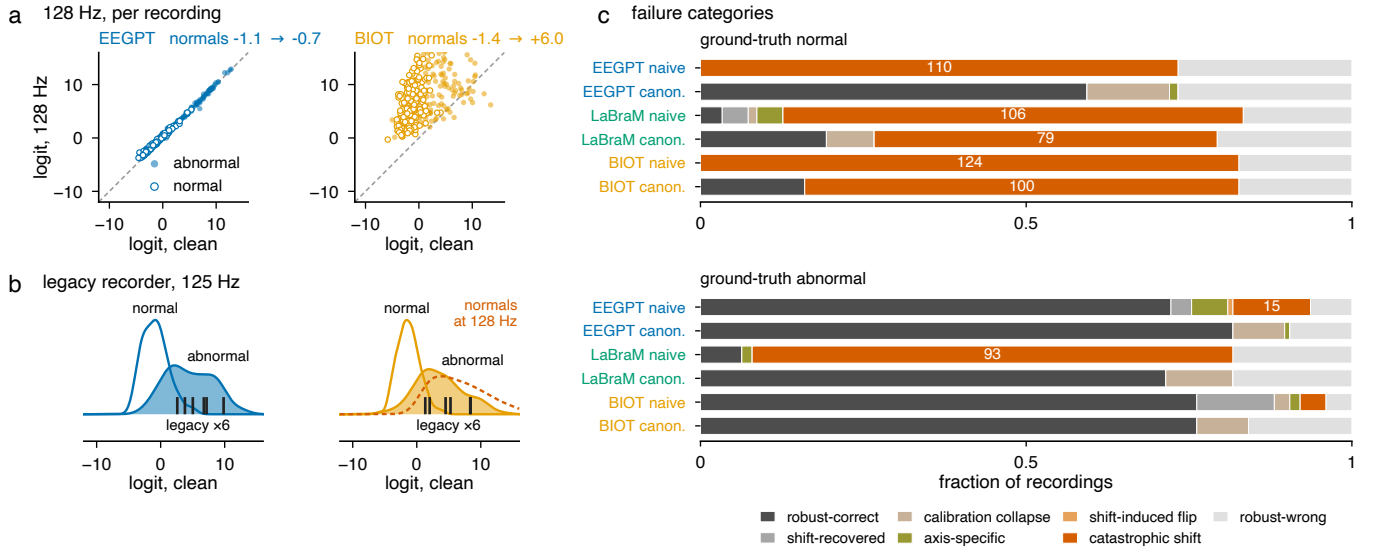


Fig. 3: **What the envelope means for a recording, a population and a recorder.** (a) Per-recording logit after 128 Hz acquisition and the full contract against the clean logit (seed-7 probes): EEGPT stays on the identity, BIOT lifts every normal recording by about 7 logits. (b) Clean eval logit distributions with the six legacy Natus recordings (125 Hz) as ticks; for BIOT the dashed curve is where its eval normals land at this rate. (c) Failure-category composition over the 12 non-identity cells of A1–A3, by backbone, arm and ground truth (counts in Table S5).

add an explicit 0.5–50 Hz band-limit stage, as BRIDGE-EEG’s preprocessing does [13]; for backbones of that bandwidth it makes every native rate from 100 Hz upward on-contract by construction. For BIOT the same stage would move the envelope edge, not remove it: a stage must be measured per backbone, not assumed.

Scope. One binary task on one public corpus, three frozen backbones, small probes, operators with known left-inverses, and operator models rather than a sample of devices. The legacy vignette is six unlabelled recordings. LaBraM’s pre-training includes other TUH subcorpora (§III-C). No clinical-efficacy claim is made; adjudicated multi-recorder validation is the next study (Appendix E).

VI. CONCLUSION

A preprocessing contract can be measured rather than assumed. Five operator-modelled shifts upstream of ESF, scored on three frozen backbones with each corrective stage on and off, give per backbone the shifts the contract closes, the stage responsible and the calibration cost. Uncorrected shift produces confident wrong reports whose direction depends on the backbone; the contract removes them where it is measured to, and nowhere else.

REPRODUCIBILITY

Code, operators, adapters, all probes, feature caches, per-cell predictions for every (backbone, probe, cell), tables and proofs: <https://github.com/h5371h/esf-recoverability> (MIT; tag v2.0.0-preprint-v2, archived on Zenodo; the withdrawn v1 snapshot remains at v1.0.0-spm2026-submission). TUAB is obtained under the NEDC agreement [5]; the legacy recordings are

clinical data, processed locally and not released. Checkpoint revisions, library versions and seeds: Appendix E. Every number regenerates from the released caches without a GPU. *Cost.* On a 30-core CPU the full contract takes 0.64 s per 25-min recording (notch 0.28, EDF read 0.23, robust z 0.17, high-pass 0.16, CAR 0.02, resample < 0.01 s); the backbones take 12 (EEGPT), 1.6 (LaBraM) and 0.5 s (BIOT), or 113/10/3 s single-threaded. The contract is never the bottleneck.

VERSION NOTE (V2)

v2 corrects v1 in six respects, all found by the author after submission: v1 applied operators *after* canonicalization (its rate axis was a time-warp, its gain arm never composed forward and inverse, its mains amplitudes were in robust-SD units); its probe was trained on a split that overlapped eval; Theorem 1 was stated marginally; $C_{\max}$ was fitted, not validated; the conflict-of-interest statement was incomplete; the channel-projection adapter was unseeded. *Withdrawn:* the v1 losses at 128 and 512 Hz for EEGPT, the montage calibration recovery, the empirical support for Proposition 2. *Added:* LaBraM, BIOT, seeds, a linear probe, bandwidth, envelope, controls, the legacy recorder. Details: Appendix E.

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and the author has a commercial interest in its adoption. The reference implementation is MIT-licensed.

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